TZOFI KLINGHOFFER

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EDUCATION

PhD, Media Arts and Sciences **Sept. 2021 – June 2027 (expected)**
Massachusetts Institute of Technology, Media Lab, Cambridge, MA
Research focus: artificial intelligence, computer vision, and sensing–perception co-design.
Advisor: Ramesh Raskar

SM, Media Arts and Sciences **2023**
Massachusetts Institute of Technology, Media Lab, Cambridge, MA
Thesis committee: Ramesh Raskar, Phillip Isola, Sanja Fidler

BS, Computer Science, *summa cum laude* **2018**
The University of Alabama, College of Engineering, Tuscaloosa, AL
Minors: Chinese; Social Innovation and Leadership. Certificate in Global Studies

PUBLICATIONS

** Equal contribution. † Student advised or mentored by applicant.*

Lead-Author Conference Papers

[1] **T. Klinghoffer**, S. Somasundaram*, X. Xiang*, Y. Fan, C. Richardt, A. Dave, R. Raskar, R. Ranjan. "Shoot-Bounce-3D: Single-Shot Occlusion-Aware 3D from Lidar by Decomposing Two-Bounce Light." ACM SIGGRAPH Asia, 2025. [Website]

[2] **T. Klinghoffer**, X. Xiang*, S. Somasundaram*, Y. Fan, C. Richardt, R. Raskar, R. Ranjan. "PlatoNeRF: 3D Reconstruction in Plato's Cave via Single-View Two-Bounce Lidar." CVPR, 2024. **Oral; Best Paper Award Finalist (top ~0.2% of submissions).** [Website] [MIT News]

[3] **T. Klinghoffer***, K. Tiwary*, N. Behari, B. Agrawalla†, R. Raskar. "DISeR: Designing Imaging Systems with Reinforcement Learning." IEEE/CVF International Conference on Computer Vision (ICCV), 2023. [Website]

[4] **T. Klinghoffer**, J. Philion, W. Chen, O. Litany, Z. Gojcic, J. Joo, R. Raskar, S. Fidler, J. Alvarez. "Towards Viewpoint Robustness in Bird's Eye View Segmentation." IEEE/CVF International Conference on Computer Vision (ICCV), 2023. [Website] [NVIDIA Technical Blog] [DRIVE Labs Episode]

[5] K. Tiwary*, **T. Klinghoffer***, R. Raskar. "Towards Learning Neural Representations from Shadows." European Conference on Computer Vision (ECCV), 2022. [Website]

[6] **T. Klinghoffer***, S. Somasundaram*, K. Tiwary*, R. Raskar. "Physics vs. Learned Priors: Rethinking Camera and Algorithm Design for Task-Specific Imaging." IEEE International Conference on Computational Photography (ICCP), 2022. [Paper] [Talk]

Other Conference Papers

[7] N. Behari, A. Young, **T. Klinghoffer**, A. Dave, R. Raskar. "Task-Driven Implicit Representations for Automated Design of LiDAR Systems." IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. **Highlight.** [Website]

[8] N. Behari, A. Young, S. Somasundaram, **T. Klinghoffer**, A. Dave, R. Raskar. "Blurred LiDAR for Sharper 3D: Robust Handheld 3D Scanning with Diffuse LiDAR and RGB." IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. **Highlight.** [Website]

[9] D. Gilo†, **T. Klinghoffer**, O. Litany. "EPI-NAF: Enhancing Neural Attenuation Fields for Limited-Angle CT with Epipolar Consistency Conditions." IEEE International Symposium on Biomedical Imaging (ISBI), 2025. **Oral.** [Paper]

[10] K. Tiwary, A. Dave, N. Behari, **T. Klinghoffer**, A. Veeraraghavan, R. Raskar. "ORCa: Glossy Objects as Radiance-Field Cameras." IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023. [Website]

[11] L. Gjesteby, **T. Klinghoffer**, M. Ash, M. Melton, K. Otto, D. Lamb, S. Burke, L. Brattain. "Annotation-Efficient 3D U-Nets for Brain Plasticity Network Mapping." IEEE International Symposium on Biomedical Imaging (ISBI), 2021.

Journal and Other Publications

[12] K. Tiwary*, A. Young*, Z. Tasneem, **T. Klinghoffer**, A. Dave, T. Poggio, D.E. Nilsson, B. Cheung*, R. Raskar*. "What if Eye…? Computationally Recreating Vision Evolution." *Science Advances*, 2025. [Website]

[13] K. Tiwary, **T. Klinghoffer***, A. Young*, S. Somasundaram, N. Behari, A. Dave, B. Cheung, D.E. Nilsson, T. Poggio, R. Raskar. "A Roadmap for Generative Design of Visual Intelligence." MIT Press, 2024. [Paper]

Workshop Papers

[14] **T. Klinghoffer***, K. Tiwary*, A. Balata, V. Sharma, R. Raskar. "Physically Disentangled Representations." ECCV Workshops, 2022. [Paper]

[15] **T. Klinghoffer**, P. Morales, Y.G. Park, N. Evans, K. Cheung, L. Brattain. "Self-Supervised Feature Extraction for 3D Axon Segmentation." IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2020.

[16] P. Morales*, **T. Klinghoffer***, S.J. Lee. "Feature Forwarding for Efficient Single Image Dehazing." IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2019.

[17] C. Ancuti, et al. "NTIRE 2019 Image Dehazing Challenge Report." IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2019.

[18] **T. Klinghoffer**, C. Perez, R. Vincent, P. Perdikaris, C. Chryssostomidis. "Applying Image Recognition to Enhance Fisheries Management Capabilities." American Meteorological Society, 17th Conference on Artificial and Computational Intelligence and its Applications to Environmental Sciences, 2018. **Student Research Award.**

Under Review

[a] E. Underwood*†, **T. Klinghoffer***, L. Collins, R. Raskar. "Coloring Plato's Cave: Generative Single-View 3D Reconstruction from RGB and Two-Bounce Lidar." Under review, 2026.

[b] **T. Klinghoffer**, S. Somasundaram, P. Liang, R. Raskar. "PhotonJEPA: Self-Supervised Learning in Low Light with Single-Photon Cameras." Under review, 2026.

[c] **T. Klinghoffer***, M. Keidar*, T. Borreda, O. Litany. "RadarDream: Joint RGB–Radar Driving Video Generation Across Visibility Conditions." Under review, 2026.

[d] N. Sortur†, J. Goodwin, P. Patel, L.E. Martinez Jr., **T. Klinghoffer**, R. Caceres, R. Walters. "Radar2Shape: 3D Shape Reconstruction from High-Frequency Radar Using Multiresolution Signed Distance Functions." Under review.

HONORS AND AWARDS

Best Paper Award Finalist, CVPR (PlatoNeRF)	**2024**
DoD National Defense Science and Engineering Graduate (NDSEG) Fellowship	**2024**
Qualcomm Innovation Fellowship	**2023**
Draper Scholar Award	**2023**
NSF Graduate Research Fellowship Program, Honorable Mention	**2023**
MIT Lincoln Scholar Award	**2020**
DoD SMART Scholar Award	**2019**
Student Research Award, American Meteorological Society	**2018**
NOAA Ernest F. Hollings Scholarship	**2016**

RESEARCH FUNDING

Hyundai America Technical Center, Inc. – $250K **June 2024 – May 2025**
Lead proposal writer (PI: Ramesh Raskar)

MISTI MIT-Israel Zuckerman STEM Fund – $30K **May 2023 – Present**
Lead proposal writer (PIs: Ramesh Raskar, Or Litany)

Advanced Concepts Committee, MIT Lincoln Laboratory – $210K **Oct. 2019 – Sept. 2020**
Lead proposal writer (PI: Peter Morales)

PATENTS

T. Klinghoffer, J. Philion, Z. Gojcic, S. Fidler, O. Litany, W. Chen, J.A.M. Lopez. "Synthetic Data Generation Using Viewpoint Augmentation for Autonomous Systems and Applications." Published U.S. Patent Application US 2024/0362897 A1, Oct. 2024.

TEACHING AND MENTORING

The University of Alabama Honors College **Jan. – May 2018**
Instructor, Programming for Non-Programmers *Tuscaloosa, AL*

- Designed and taught a new undergraduate-level, project-based introduction to programming course

BlendEd – *Guest lecturer and mentor* **May 2026 – Present**

Undergraduate research mentorship

Eli Underwood	Feb. 2026 – Present	lead author on [a]; now applying to PhD programs
Nilay Mishra	Jan. – May 2025	MIT undergraduate
Bhavya Agrawalla	2022 – 2024	co-author on [3]; now PhD student at CMU
Dewei Feng	2022 – 2023	now Software Engineer, Meta
Mimi Lohanimit	2021 – 2022	now Software Engineer, Applied Intuition

Graduate admissions mentorship – mentored Ishita Kakkar (now PhD student, University of Wisconsin; NSF GRFP recipient); Volunteer with MIT Media Lab SOS program

ACADEMIC SERVICE AND OUTREACH

Lead Organizer: Workshop on Neural Fields Beyond Conventional Cameras (ECCV 2024, CVPR 2025)

Organizer: MIT Media Lab workshops on multimodal AI for extreme sensing and AI for scientific discovery

Reviewer

- Conferences: CVPR, ICCV, ECCV, ICML, NeurIPS, ICRA
- Journals: IEEE TPAMI, ACM TOG, IEEE TCI

Outreach: FIRST LEGO League coaching; MIT Beaver Works

INDUSTRY EXPERIENCE

Meta Reality Labs **May – Sept. 2023, 2024, 2025**
AI Research Scientist Intern: 3D vision and vision-language models for extended reality *Cambridge, MA*

Code Metal **Dec. 2023 – Present**
Part-Time Founding Researcher: edge computer vision and hardware–software co-design *Boston, MA*

NVIDIA Research **May 2022 – Jan. 2023**
Research Intern: neural rendering for autonomous-vehicle perception *Remote*

Amazon **Aug. 2020 – Sept. 2021**
Software Development Engineer II, Alexa AI *Cambridge, MA*

- Led design and implementation of software for automated generation of training and test datasets

MIT Lincoln Laboratory **May 2018 – Aug. 2020**
Associate Technical Staff, Homeland Protection Group (Secret clearance) *Lexington, MA*

- Developed machine learning and computer vision methods for national security mission areas
- In collaboration with MIT CSAIL, led computer vision research on segmentation and classification of pathologies in medical images, including x-ray and microscopy, resulting in 3 publications
- Contributed to development and deployment of real-time software systems that improved anomaly detection for critical areas of homeland security by over 600%

Earlier internships: MIT Sea Grant, research intern, object detection for NOAA fisheries management (2017); Lockheed Martin Space Systems, software engineering intern, Orion mission (2016); Jacobs Technology, software development and test intern, U.S. Air Force systems (2014, 2015)

INVITED TALKS

MIT CSAIL Visual Computing Seminar – “Learning from Sensors, Learning for Sensors”	**Oct. 2026**
Rising Stars, MIT Media Lab – “What If AI Could See Hidden Objects?”	**Oct. 2025**
Carnegie Mellon University, Computational Imaging Groups – “Seeing Hidden Objects”	**June 2024**
Technion – Israel Institute of Technology, LIT Lab – “Seeing Hidden Objects”	**May 2024**
Hyundai Vision Conference – “Imaging through Shadows and Reflections”	**Aug. 2023**
Recent Advances in AI for National Security – “Dense Axon Segmentation”	**Nov. 2019**

MEDIA COVERAGE

“What if Eye…? Computationally Recreating Vision Evolution” **2025**
Featured in MIT News, Hacker News, MSN, and more

“PlatoNeRF: 3D Reconstruction in Plato’s Cave via Single-View Two-Bounce Lidar” **2024**
Featured in MIT News, MarkTechPost, ScienceDaily, Optics.org, and more

“ORCa: Glossy Objects as Radiance-Field Cameras” **2023**
MIT front-page spotlight; featured in SciTechDaily, MarkTechPost, and more